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 Studierichting: Informatica
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$$\begin{aligned}
 & \lim_{\lambda \rightarrow 0} \frac{f\left(\left(\frac{\pi}{3}, \frac{-2\pi}{3}\right) + \lambda(4, -3)\right) - f\left(\frac{\pi}{3}, \frac{-2\pi}{3}\right)}{\lambda} \\
 &= \lim_{\lambda \rightarrow 0} \frac{f\left(\frac{\pi}{3} + 4\lambda, \frac{-2\pi}{3} - 3\lambda\right) - f\left(\frac{\pi}{3}, \frac{-2\pi}{3}\right)}{\lambda} \\
 &= \lim_{\lambda \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4\lambda\right) \cdot \cos\left(\frac{-2\pi}{3} - 3\lambda\right) - \sin\frac{\pi}{3} \cdot \cos\frac{-2\pi}{3}}{\lambda} \\
 &= \lim_{\lambda \rightarrow 0} \frac{-(-\sqrt{3} \cdot 4 \sin(4\lambda) + 4 \cdot \cos(4\lambda))(\cos(3\lambda) + \sqrt{3} \sin(3\lambda)) + (-3 \sin(3\lambda) + 3\sqrt{3} \cos(3\lambda))(\sqrt{3} \cos(4\lambda) + \sin(4\lambda))}{4} \\
 &= -\frac{13}{4}
 \end{aligned}$$

References